

Revised May 7, 2026 – Final LRBAAsubmission Version

Property of Fratres X AI
GitHub: <https://github.com/FratresMedAI/Ancile-Aeris>

Ancile Aeris v2.0 Quad Chart

LRBAAsubmission Counter-UAS / Anti-Terror Defensive System Property of Fratres X AI GitHub:
<https://github.com/FratresMedAI/Ancile-Aeris>

Quad Chart Layout

Use a four-quadrant visual layout for PDF conversion:

Technical Approach	Innovation
ROS 2 modular C-UAS stack with sensing, fusion, clean DARKSPACE audit, PID safety gates, mesh scout ISR, cognitive EW, and non-kinetic effectors.	Full layered non-kinetic defeat stack with intelligent cognitive selection and XAI rationale.
Team	Impact
Fratres X AI: software-defined defense prototype team delivering buildable ROS 2 packages, GitHub verification, and submission-ready artifacts.	Human-governed, auditable, low-collateral C-UAS response model aligned to DHS and modular acquisition pathways.

Quadrant 1 - Technical Approach

Ancile Aeris v2.0 is a working ROS 2 C-UAS platform that demonstrates:

- ? Multi-modal simulated sensing: visual, thermal, acoustic, RF, lidar, and SIGINT-style inputs.
- ? Deterministic fusion through /fused_tracks with PID >= 0.999 safety context.
- ? Mesh scout mothership ISR through /scout_eyes and /mesh/mothership_swarm_status.
- ? DARKSPACE audit trail through /audit/events and /darkspace/status.
- ? Non-kinetic effector selection through /effector/selected_plan and /effector/status.
- ? Cognitive EW output through /cognitive_ew_commands.
- ? Human-on-the-loop safety gates and monitor-only behavior unless authorization is satisfied.

Suggested visual: left-to-right pipeline:

Sensors -> Fusion -> PID Safety Gate -> Mesh Scout ISR -> Effectors -> Cognitive EW -> DARKSPACE/XAI

Quadrant 2 - Innovation

The key differentiator is the layered non-kinetic defeat stack:

Mode	Defensive Role
monitor	Default passive ISR and no-action state

Mode	Defensive Role
multi_sensor_deception	Deception and adversary sensor confusion modeling
cognitive_jamming	Adaptive RF denial recommendation
gnss_link_spoofing	Navigation / link spoofing concept modeling
hpm_denial_stub	HPM-class denial concept, simulation-only
control_link_takeover	Authorized recovered-link takeover concept, dual-auth gated

Ancile Aeris v2.0 selects among these modes using:

```
agent_orchestrator -> digital_twin -> cognitive_ew
```

The recommendation is published with selected mode, family, authorization state, monitor-only status, and XAI rationale. This creates a credible, lethally effective defensive story while preserving lawful human authority.

Suggested visual: effector catalog wheel with cognitive EW at center and DARKSPACE/XAI as the assurance layer.

Quadrant 3 - Team

Fratres X AI has delivered Ancile Aeris v2.0 as a working software-defined defense prototype:

- ? GitHub implementation: <https://github.com/FratresMedAI/Ancile-Aeris>
- ? Buildable ROS 2 packages across sensing, fusion, safety, audit, scout ISR, cognitive EW, effectors, and bringup.
- ? Verified final build: `colcon build --symlink-install --packages-up-to ancile_aeris_bringup`
- ? Verified final launch: `ros2 launch ancile_aeris_bringup ancile_aeris_basic_demo.launch.py`
- ? Verified DARKSPACE: `integrity_ok: true, chain_gap_count: 0`, zero hash-chain mismatch warnings.

Suggested visual: GitHub repository screenshot, package tree, and final clean build terminal output.

Quadrant 4 - Impact

Impact Area	Value
Operational	Faster track correlation, mesh ISR, and layered response recommendations for event and infrastructure protection.
Non-Kinetic Defeat	HPM-class denial, cognitive jamming, GNSS/link spoofing, deception, and authorized takeover modeled as explainable options.
Oversight	DARKSPACE audit status, /audit/events, XAI rationale, and human-on-loop gates.
Acquisition	Modular alignment with JIATF-401 Marketplace and Replicator 2 rapid-fielding concepts.
Interoperability	Lattice-style mesh awareness, DroneHunter F700-class capture narrative, Leonidas-style non-kinetic framing, and DHS PEO UAS/C-UAS mission alignment.

Suggested visual: "Verified Final Demo" callout:

```
/darkspace/status -> integrity_ok: true, chain_gap_count: 0  
/effector/selected_plan -> selected mode + XAI rationale  
/cognitive_ew_commands -> xai_rationale + monitor_only
```

Government / Industry Alignment

Ancile Aeris v2.0 aligns with Anduril Lattice-style mesh awareness, JIATF-401 Marketplace integration, Fortem DroneHunter F700-class low-collateral response concepts, Epirus Leonidas-like HPM non-kinetic framing, Replicator 2 rapid fielding, and DHS Program Executive Office for UAS/C-UAS defensive mission needs.

Simulation-only C-UAS defensive demonstration. No autonomous weapon release. Government and industry references are alignment examples only and do not imply endorsement.

Production Notes

- ? Convert into a one-page quad chart with four equal quadrants.
- ? Use high-contrast defense-tech styling: dark navy, white, steel gray, and one accent color.
- ? Add icons for sensor fusion, mesh ISR, non-kinetic effectors, DARKSPACE audit, and human authorization.
- ? Include QR code to the GitHub repository and footer: "Ancile Aeris v2.0 - Property of Fratres X AI."

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